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TRANSFORMERS

Principles • Construction • Applications • Troubleshooting

A Course Guide by AYE Tech Hub

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1. Introduction

Transformers are the backbone of every modern electrical power system. From the moment electricity leaves a generator to the instant it powers a robot cell on the factory floor, it passes through at least three or four transformers — each one silently adjusting voltage and current to suit the next stage of the journey. Understanding how they work, how they are built, and how to keep them healthy is a core competency for any industrial engineer or maintenance technician.

This AYE Tech Hub guide delivers a senior-engineer-level walkthrough of transformers across ten sections: the physics of electromagnetic induction, the anatomy of a power transformer, step-up and step-down operation, three-phase vector groups, losses and cooling, real factory applications, and a structured troubleshooting routine you can use immediately on the shop floor.

Learning Objectives

By the end of this guide you will be able to:

- Explain Faraday's law and derive the EMF equation for a transformer.
- Identify every major component of a power transformer and state its function.
- Calculate turns ratio, voltage ratio, current ratio and efficiency for any transformer.
- Recognise Δ -Y, Y- Δ and Y-Y connections and select the correct vector group.
- Select the correct cooling class (ONAN, ONAF, OFAF) for a given rating and load profile.
- Troubleshoot common transformer faults using IR, TTR, DGA, and thermographic tests.

2. Electromagnetic Induction — The Working Principle

A transformer is a **static electromagnetic device** — no moving parts, no direct electrical connection between input and output — yet it transfers power with efficiencies routinely above 98 %. This performance rests on two laws of physics: **Faraday's Law of Electromagnetic Induction** and **Ampere's Law**.

When alternating current flows through the primary winding, it creates an alternating magnetic flux Φ in the iron core. That alternating flux threads the secondary winding and — by Faraday's law — induces an EMF in

it. The magnitude depends on the rate of change of flux and the number of secondary turns.

Faraday's Law of Electromagnetic Induction

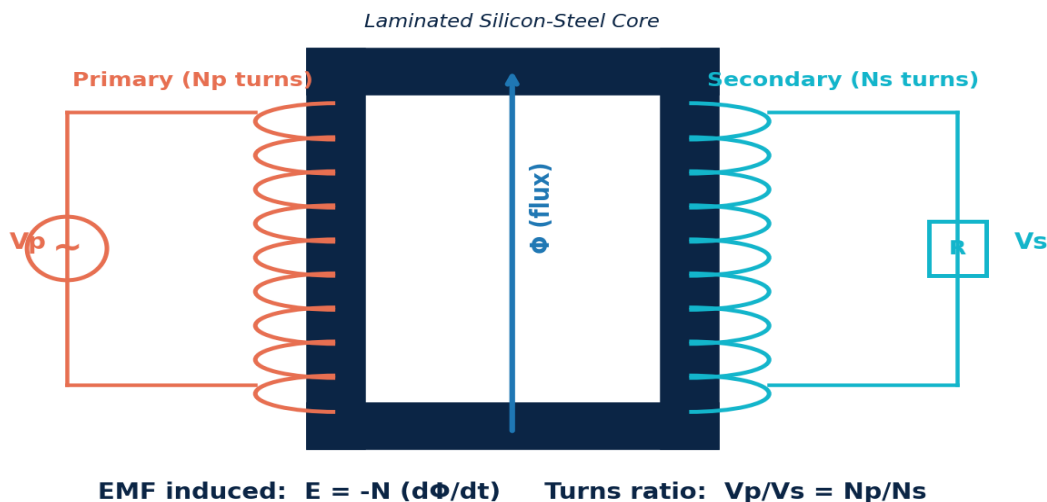


Figure 2.1 — AC current in the primary creates an alternating flux Φ in the laminated core, which induces a voltage in the secondary winding.

Key Equations

Faraday's induced EMF: $E = -N \cdot (d\Phi/dt)$

Turns ratio (a): $a = N_p/N_s = V_p/V_s = I_s/I_p$

EMF equation: $E = 4.44 \cdot f \cdot N \cdot B_{\max} \cdot A_c$

Power conservation (ideal): $V_p \cdot I_p = V_s \cdot I_s$

Where f = supply frequency (50 Hz in Korea/Ethiopia), N = number of turns, $B_{\max}$ = peak flux density (1.4–1.7 T for CRGO silicon steel), and A_c = core cross-sectional area in m^2 .

Engineer's Insight

Step up the voltage and the current drops in exact proportion — $V_p \cdot I_p = V_s \cdot I_s$. This is why transmission lines run at 154 kV or 345 kV: at high voltage the current is tiny, I^2R losses in the cables are negligible, and you can transmit hundreds of megawatts over hundreds of kilometres with minimal energy waste.

Worked Example — EMF and Turns Calculation

A single-phase transformer has a primary voltage of 230 V at 50 Hz. The core has a cross-sectional area of 0.004 m^2 and a maximum flux density $B_{\max} = 1.5$ T. Using the EMF equation $E = 4.44 \cdot f \cdot N \cdot B_{\max} \cdot A_c$:

Step	Calculation	Result
Primary turns	$N_p = V_p / (4.44 \times f \times B_{\max} \times A_c)$	$N_p = 230 / (4.44 \times 50 \times 1.5 \times 0.004) = 172$ turns
Turns ratio for 11 kV secondary	$a = V_p / V_s = 230 / 11\,000$	$a = 0.0209 \rightarrow N_s = N_p / a = 8\,229$ turns
Primary current at 50 kVA	$I_p = S / V_p = 50\,000 / 230$	$I_p = 217$ A
Secondary current	$I_s = S / V_s = 50\,000 / 11\,000$	$I_s = 4.55$ A
Check: $I_p \times V_p = I_s \times V_s$	$217 \times 230 = 4.55 \times 11\,000$	49 910 W $\approx$ 50 050 W ✓ (ideal TX)

The EMF equation is central to core design. A designer choosing a larger core area (A_c) needs fewer turns for the same voltage — reducing copper cost and winding resistance. A smaller core needs more turns — increasing copper weight and I^2R losses. The operating point on the B-H curve must stay below saturation (typically 1.7 T for CRGO steel) to avoid a large magnetising current surge.

3. Transformer Construction & Components

An industrial power transformer is built around four functional groups: the **magnetic circuit** (core), the **electrical circuit** (windings), the **insulation system** (oil and solid insulation), and the **auxiliary system** (tank, bushings, conservator, protection devices). Modern distribution transformers routinely last 30–40 years in service.

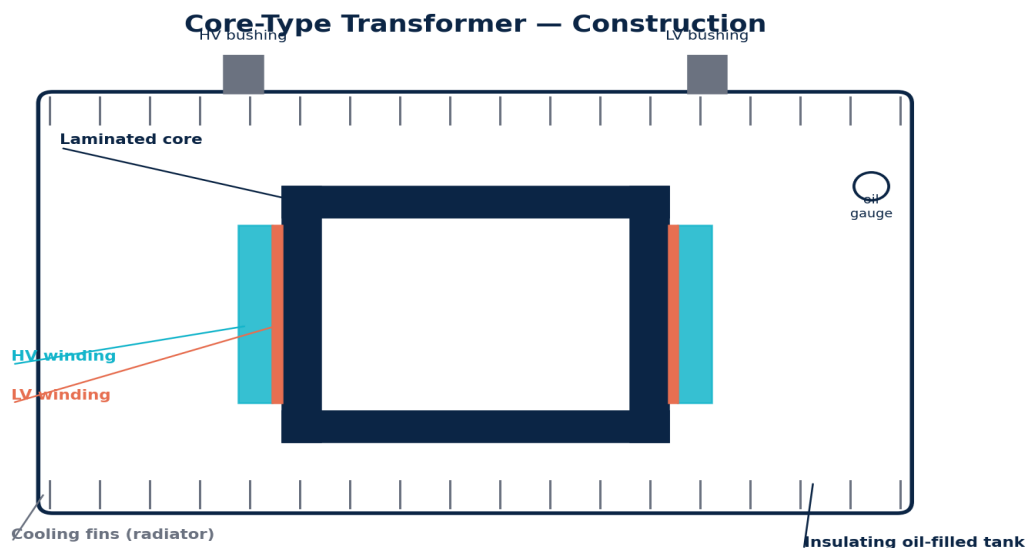


Figure 3.1 — Cutaway view of a core-type oil-immersed distribution transformer.

Component	Material / Standard	Function
Laminated core	CRGO silicon steel, 0.27–0.35 mm	Low-reluctance magnetic path; laminations cut eddy-current losses by >90%.
HV winding	Enamelled copper wire/foil	Many turns of thin conductor — high voltage, low current side.
LV winding	Enamelled copper bar/foil	Fewer turns of heavy conductor — low voltage, high current side.
Insulating oil	Mineral or ester oil	Dielectric insulator and coolant; convects heat to tank wall.
Solid insulation	Kraft paper, pressboard	Inter-winding and inter-layer barrier; oil-impregnated.
Bushings	Porcelain or silicone polymer	Sealed HV/LV feed-throughs out of the oil-filled tank.
Conservator	Welded steel cylinder	Expansion reservoir; keeps oil level constant with temperature.
Buchholz relay	Gas-and-oil relay (IEC 60214)	Detects internal arcing via gas accumulation — first protection.
Tap-changer	On-load (OLTC) or off-circuit	Adjusts turns ratio $\pm 2.5\%/\pm 5\%$ to regulate secondary voltage.
Radiator fins	Corrugated or panel type	Extend cooling area; up to 30% more dissipation vs plain tank.

4. Step-Up and Step-Down Transformers

The terms **step-up** and **step-down** describe which direction voltage moves between primary and secondary. Mechanically the two types are identical — same core, same oil, same tank. Only the turns ratio and which

terminal is energised changes.

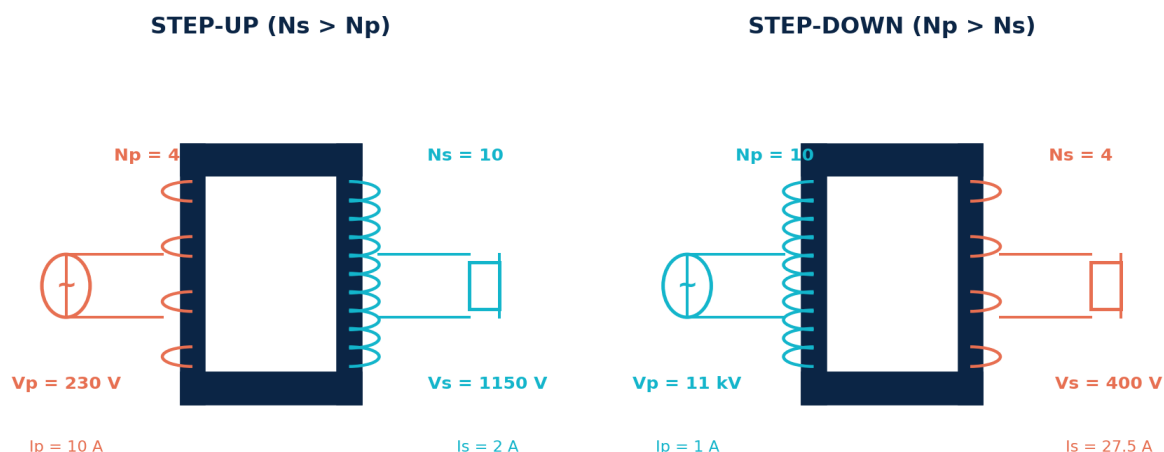


Figure 4.1 — Step-up (left): $N_s > N_p$, voltage rises, current falls. Step-down (right): $N_p > N_s$, voltage falls, current rises. Power is conserved.

Step-Up Transformers

Used at the generator output (e.g., 11 kV $\rightarrow$ 154 kV) for long-distance transmission. Industrial applications include **induction furnaces**, **X-ray generators**, **electrostatic precipitators** and high-voltage testing laboratories.

Step-Down Transformers

Far more numerous on the factory floor. They reduce utility voltage (22.9 kV in Korea, 15 kV in Ethiopia) to 380 V for motors and drives, plus **control transformers** (100 VA–5 kVA) that step it further to 220 V/110 V/24 V for PLC cards and solenoids.

Parameter	Formula	Example — 500 kVA, 22.9 kV / 380 V
Turns ratio (a)	$a = N_p/N_s = V_p/V_s$	$a = 22\,900 / 380 = 60.3$
Rated HV current	$I_p = S / (\sqrt{3} \times V_p)$	$500\,000 / (1.732 \times 22\,900) = 12.6\text{ A}$
Rated LV current	$I_s = S / (\sqrt{3} \times V_s)$	$500\,000 / (1.732 \times 380) = 759\text{ A}$
Impedance voltage	$V_z\% = (V_{sc}/V_n) \times 100$	Typically 4–6 % for distribution TX
Short-circuit current	$I_{sc} = I_n / (V_z\%/100)$	$759 / 0.05 = 15\,180\text{ A}$ (at 5% V_z)

Where Transformers Sit in the Power Grid

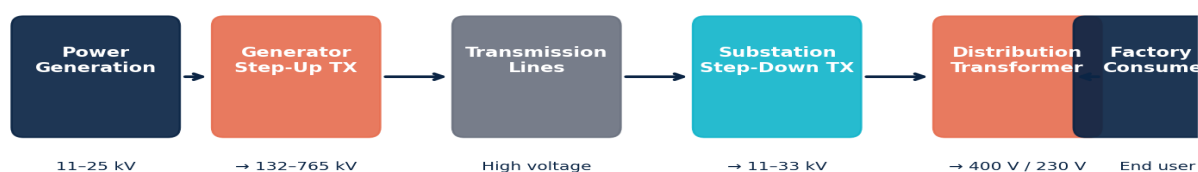


Figure 4.2 — Voltage transformation cascade from power generation to the factory consumer.

5. Transformers in the Power Grid

Electricity travels from generator to consumer through a cascade of voltage transformations. Each step is optimised for a different requirement — high voltage for long-distance efficiency, medium voltage for urban distribution, low voltage for safe end-use. Transformers sit at every junction in this chain.

Stage	Typical Voltage	TX Type	Purpose
Generation (alternator)	11–25 kV	—	Generator terminal voltage
Transmission step-up	132–765 kV	Generator TX (GSU)	Reduce I^2R losses over long lines
Transmission	154–765 kV	—	National grid backbone
Sub-transmission step-down	22–66 kV	Grid substation TX	Feed major industrial zones
Distribution step-down	380 V / 220 V	Pole / pad-mounted TX	Supply factories & buildings
Control / instrument	24 V / 110 V	Control TX, CT, PT	PLC, relay & metering circuits

Instrument Transformers — CT and PT

Two special transformer types serve metering and protection, not load power. A **Current Transformer (CT)** steps down primary current (e.g., 1 000 A) to a safe 5 A secondary for ammeters, energy meters and overcurrent relays. A **Potential Transformer (PT/VT)** steps down high voltage (e.g., 22.9 kV) to a standard 110 V secondary for voltmeters and distance relays. Two absolute rules: **never open-circuit a CT secondary** (produces dangerous high voltage), and **never short-circuit a PT secondary** (produces destructive current).

Type	Ratio	Secondary Standard	Key Safety Rule
Current TX (CT)	5–5000 A / 5 A or 1 A	IEC 61869-2	NEVER open the secondary while primary is energised
Potential TX (PT)	Up to 33 kV / 110 V	IEC 61869-3	NEVER short-circuit the secondary
Earthing TX	Phase-neutral / low Z	IEC 60076	Provides neutral point for isolated systems

Field Note

In Korea the standard distribution voltage is **22.9 kV (KEPCO)**. Factory main transformers are rated 22.9 kV / 380-220 V, Dyn11. In Ethiopia the standard is **15 kV or 33 kV** depending on region. Always confirm the local utility voltage before ordering a replacement transformer — the wrong HV tap is a costly error that can take weeks to rectify.

6. Three-Phase Transformers & Vector Groups

Industrial power systems are almost entirely three-phase. A three-phase transformer can be built as a bank of three single-phase units or — far more commonly — as a single unit with a three-legged core. The single three-phase unit is cheaper, smaller, lighter, and more efficient.

Each winding set (primary and secondary) can be connected in **Star (Y)**, **Delta (Δ)**, or **Zigzag (Z)**. The combination determines voltage ratio, neutral availability, and the phase angle shift between HV and LV sides — described by the **vector group**.

Three-Phase Transformer — Delta-Wye (Dy11) Connection

- HV side: 3 line wires, no neutral
- LV side: 4 wires (3-phase + N)
- 30° phase shift between HV & LV (vector group 11)
- Most common for distribution

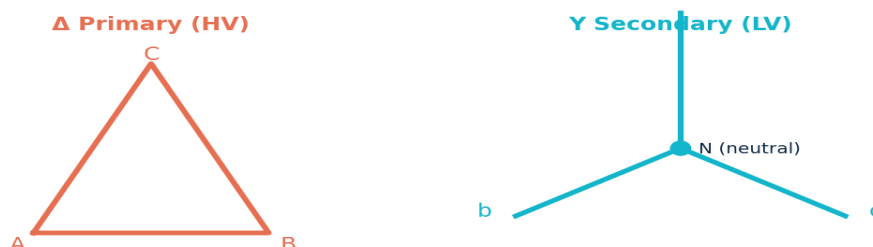


Figure 6.1 — Dyn11: Delta primary (HV), Star secondary (LV) with neutral. The '11' means the LV vector lags HV by 330° (equivalent to 30° lead) — like the 11 o'clock position on a clock face.

Vector Group	HV / LV Connection	Phase Shift	Neutral	Typical Application
Dyn11	Delta / Star with neutral	30° lag	LV only	Most common distribution (22.9 kV → 380 V)
YNd1	Star with neutral / Delta	30° lead	HV only	Generator step-up, large industrial plants
YNyn0	Star / Star, both neutrals	0°	Both	Auto-grounded networks, instrument TX
Dd0	Delta / Delta	0°	None	Industrial furnace, high harmonic loads
Yzn11	Star / Zigzag + neutral	30° lag	LV only	Earthing TX, unbalanced LV loads

Parallel Operation — Critical Rules

When paralleling two transformers ALL four conditions must be met simultaneously: (1) **vector groups must match** — Dyn11 cannot parallel with Dyn1; (2) **voltage ratios must be equal** at the same tap position; (3) **percentage impedances within 10%** of each other; (4) **same frequency and phase rotation**. Violating any rule causes heavy circulating currents that trip both units — a classic commissioning-day failure.

7. Losses, Efficiency & Cooling

No transformer is perfectly efficient. Losses appear as heat and must be removed by the cooling system. Understanding where losses come from lets you select the right transformer for a given load profile and diagnose overheating faults.

Loss Type	Technical Cause	How to Minimise
Hysteresis	Energy reversing magnetic domains in the core each AC cycle	High-grade CRGO silicon steel; amorphous metal cores (near-zero)
Eddy current	Circulating currents induced inside the core by changing flux	Thin laminations (0.27–0.35 mm); varnish coating between sheets
Copper (I^2R)	Resistive heating of primary and secondary conductors	Larger conductor cross-section; lower current density design
Stray	Leakage flux losses in tank walls, clamps and structural bolts	Magnetic shunts on tank; non-magnetic stainless steel clamps
Dielectric	Polarisation losses in oil and paper/pressboard insulation	Clean, dry, degassed oil; high-grade pressboard insulation

Efficiency

Transformer efficiency: $\eta = P_{\text{out}} / (P_{\text{out}} + P_{\text{Fe}} + P_{\text{Cu}})$. Maximum efficiency occurs when iron losses equal copper losses ($P_{\text{Fe}} = P_{\text{Cu}}$), typically at 50–70% of full load. Modern IE3 / Eco-design distribution transformers achieve $\eta > 98.5\%$ at full load.

Typical Loss Distribution in a Distribution Transformer

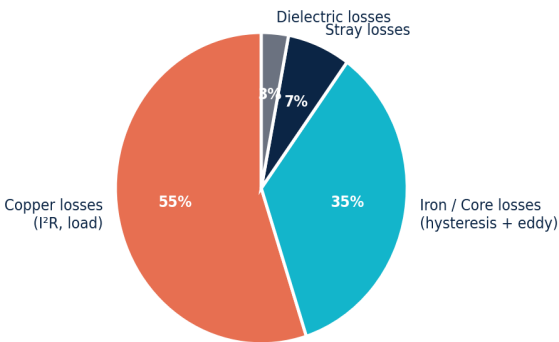


Figure 7.1 — Typical loss split in a loaded distribution transformer.

Transformer Cooling Methods — Typical Capacity

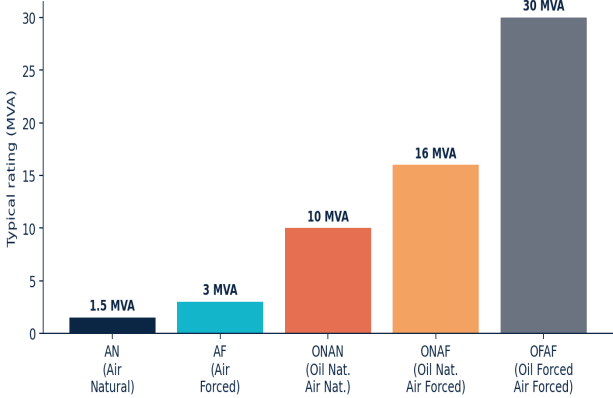


Figure 7.2 — Capacity ranges for IEC 60076 cooling classes.

Cooling Classes (IEC 60076-2)

The IEC cooling designation uses four letters: the first two describe the internal coolant (O = oil, A = air) and its circulation (N = natural, F = forced); the last two describe the external coolant similarly. **ONAN** is standard for distribution TX up to ~10 MVA. Larger units use **ONAF** (forced-air fans) or **OFAF** (forced oil + forced air) to achieve higher ratings from the same core and windings. Overloading a transformer beyond its cooling class shortens insulation life exponentially — a rule of thumb is that every 6–8 °C above rated temperature halves the insulation lifetime.

8. Industrial Applications

A modern auto-parts manufacturing plant — such as ASAN Auto Parts in Gimpo — operates a layered transformer network that delivers the right voltage to every load, from a 22 kW pressing machine to a 24 V PLC input card. Understanding this network is essential for systematic fault-finding and planned maintenance.

Factory Power Distribution at ASAN-Style Plant (Typical)

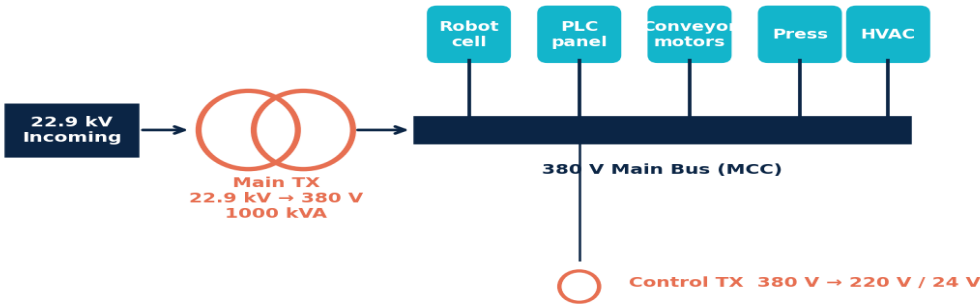


Figure 8.1 — Simplified single-line diagram of a typical auto-parts factory power distribution network showing main and control transformers.

Application	Typical Rating	Voltage	Role in the Plant
Main step-down TX	500–2500 kVA	22.9 kV → 380 V	Primary energy gateway; feeds the MCC bus
Drive isolation TX	10–200 kVA	380 V → 380 V	Galvanic isolation for VFDs; blocks conducted EMI

Application	Typical Rating	Voltage	Role in the Plant
Welding TX	20–100 kVA	380 V → 60–90 V	High secondary current for resistance/arc welding
Induction furnace TX	100–5000 kVA	380 V → variable	Supplies medium-frequency coils for heat treatment
Control TX	100 VA–5 kVA	380 V → 220/110/24 V	Powers PLC racks, HMI panels, solenoids
Current TX (CT)	5–2000 A / 5 A	N/A	Scales line current for metering & overcurrent relays
Voltage TX (PT)	Up to 33 kV / 110 V	N/A	Scales line voltage for metering & distance relays
UPS isolation TX	1–100 kVA	220 V → 220 V	Filters neutral-earth voltage in sensitive IT systems

Maintenance Field Note

On the shop floor, four warning signs demand immediate investigation: (1) a humming that suddenly increases in pitch or volume; (2) a smell of burnt varnish or hot oil near the tank; (3) a Buchholz or WTI (winding temperature indicator) alarm on the SCADA panel; (4) oil visible on the ground under the transformer. Any one of these can precede a catastrophic failure. Log the observation, reduce load if possible, and schedule an IR + DGA test within 24 hours — do not wait for the next scheduled maintenance window.

9. Troubleshooting & Preventive Maintenance

Transformer failures are almost always predictable. Industry data shows that over 80% of failures give measurable warning signs weeks or months before the unit fails. A disciplined test programme catches these signals early — before a production shutdown.

Transformer Troubleshooting — Decision Flow

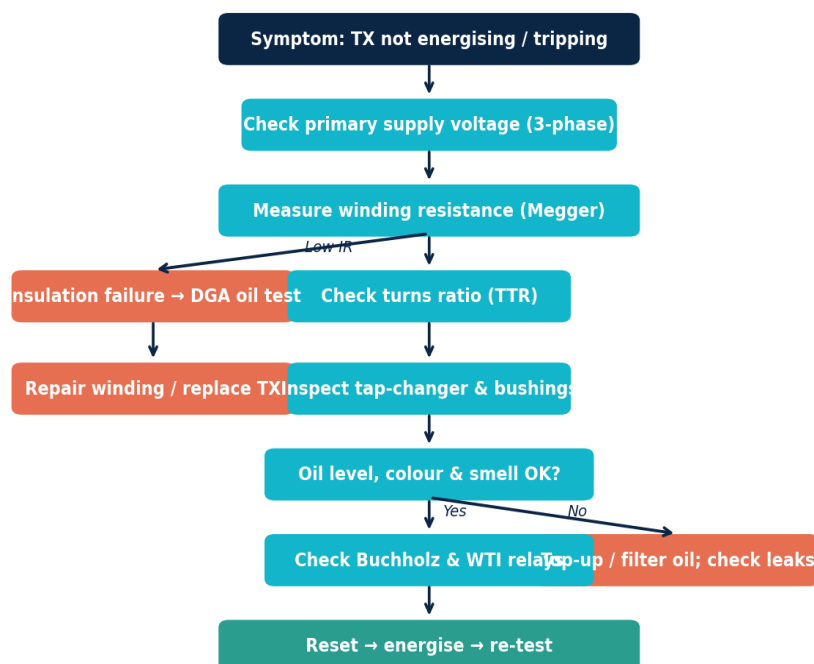


Figure 9.1 — First-pass troubleshooting decision flow for an industrial transformer that has tripped or refuses to energise.

Standard Diagnostic Tests

Test	Instrument	Pass / Acceptance Criterion
Insulation Resistance (IR)	5 kV Megger	$IR > 1 \text{ G}\Omega$ at 20°C; Polarisation Index > 2.0
Polarisation Index (PI)	Megger 10 min	$PI = IR_{10\text{min}} / IR_{1\text{min}} > 2.0$
Turns Ratio (TTR)	TTR bridge	Within $\pm 0.5\%$ of nameplate turns ratio
Winding Resistance	Micro-ohmmeter	Balanced within 2% between all three phases
Dissolved Gas Analysis (DGA)	Oil sample to lab	H_2 , C_2H_2 , CO below IEC 60599 action levels
Oil Breakdown Voltage (BDV)	BDV tester	$> 30 \text{ kV}$ across 2.5 mm gap (IEC 60156)
Oil Moisture Content	Karl Fischer	$< 15 \text{ ppm}$ for 72 kV class transformer
Thermography	FLIR / IR camera	No hotspot $> 10^\circ\text{C}$ above ambient temperature
Partial Discharge	PD analyser	$< 500 \text{ pC}$ for distribution class transformer

Safety First — LOTO Required

ALWAYS de-energise and isolate both HV and LV sides. Apply lock-out/tag-out on all disconnectors. Discharge residual charge with an earth stick. Verify dead with an approved voltage detector before touching any winding, bushing or connection. Capacitive coupling between windings can hold lethal charge for several minutes after de-energisation — do not rely on isolation alone.

Preventive Maintenance Schedule

Frequency	Task
Monthly	Visual inspection — oil level, oil colour, leaks, Buchholz relay, temperature gauges, fan operation
Quarterly	Clean bushings and HV terminals; check all bolted connections for tightness; inspect earthing bond
Annually	IR test (Megger), TTR test, oil BDV test, thermographic survey of all HV/LV terminals
3–5 Years	Full DGA oil analysis, winding resistance, moisture-in-oil test, tap-changer inspection and contact cleaning
On-demand	After any Buchholz or OTI/WTI trip: immediate DGA + IR before re-energisation

10. Summary, Key Formulas & Homework

Lesson Recap

- A transformer transfers AC energy via a shared magnetic flux — no moving parts, no direct electrical connection between primary and secondary.
- The turns ratio sets the voltage ratio; the current ratio is the exact inverse (ideal TX).
- Step-up TXs raise voltage at generation; step-down TXs lower it progressively to the load, including control TXs inside every PLC panel.
- Dyn11 is the standard vector group for 22.9 kV / 380 V factory supply in Korea; YNd1 is typical for generator step-up.
- Losses = iron (constant, load-independent) + copper (proportional to I^2). Maximum efficiency occurs when $P_{Fe} = P_{Cu}$.
- Cooling class governs the permissible continuous load. Every 6–8 °C above rated temperature halves the insulation service life.
- A structured test routine — IR, TTR, DGA, thermography — detects 80% of developing faults before they



cause a production shutdown.

Essential Formula Reference

Formula	Symbol Key	Engineering Use
$a = N_p/N_s = V_p/V_s = I_s/I_p$	a =turns ratio, N =turns, V =voltage, I =current	Any single-phase TX voltage/current calculation
$E = 4.44 \cdot f \cdot N \cdot B_{max} \cdot A_c$	f =Hz, N =turns, B_{max} =T, A_c =m ²	Core design; verify against nameplate EMF
$\eta = P_{out}/(P_{out}+P_{Fe}+P_{Cu})$	P_{Fe} =iron loss (W), P_{Cu} =copper loss (W)	Efficiency at any load level
$I_p = S/(\sqrt{3} \cdot V_p)$ (3-phase)	S =rated kVA, V_p =line voltage	Size HV fuses, cables and disconnectors
$I_s = S/(\sqrt{3} \cdot V_s)$ (3-phase)	S =rated kVA, V_s =secondary line voltage	Size LV cables, busbars and circuit breakers
$I_{sc} = I_n/(V_z\%/100)$	I_n =rated current, $V_z\%$ =impedance voltage	Fault-level calculation for switchgear selection
$\Delta T = (I/I_n)^2 \times \Delta T_n$	I =load current, ΔT_n =rated temp rise (°C)	Check thermal loading under overload condition
$PI = IR_{10min} / IR_{1min}$	IR =insulation resistance (GΩ)	Insulation quality assessment — target > 2.0

Homework Assignment

Choose any transformer in your workplace (or the nearest available plant). Document the following and submit to **awetgknway@gmail.com** with subject *Transformers Homework — Your Name*:

1. Photograph and transcribe the full nameplate (kVA, HV/LV voltage, vector group, cooling class, $V_z\%$, year of manufacture).
2. Calculate the rated HV current, rated LV current, and prospective short-circuit current.
3. Identify the cooling class and explain whether it is adequate for the current load profile.
4. List two observable warning signs you would watch for on a weekly inspection round and explain what each indicates internally.
5. Sketch the single-line diagram showing what loads the transformer feeds.

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